

OLS Regression in Excel

ECON 300 - Labour Economics

Learning Objectives

By the end of these notes, you will be able to:

- Explain what a regression is and what OLS (Ordinary Least Squares) does.
- Set up consumption and income data in Excel.
- Make a scatter plot and add a regression (trendline) in Excel.
- Run OLS using Excel's *Regression* tool (Analysis ToolPak).
- Interpret key outputs: coefficients, R^2 , and p-values.
- Compute predicted values and residuals and create a residual plot.

What is Regression? (Plain Language)

A **regression** is a method used to summarize how an outcome variable changes with a predictor variable.

Key Idea

Regression fits a line through a scatter plot of data points to describe the relationship between two variables.

In this note, we study how **Consumption** depends on **Income**.

Variables and the Model

Definitions

- **Income** (Y_i): the income for observation i (e.g., household i).
- **Consumption** (C_i): the consumption for observation i .
- **Observation** (i): one unit in the data (e.g., one household).

The Simple Linear Regression Model

We write:

$$C_i = \beta_0 + \beta_1 Y_i + u_i, \tag{1}$$

where:

- β_0 is the **intercept** (also called **autonomous consumption**).
- β_1 is the **slope** (interpreted as the **marginal propensity to consume**, MPC).
- u_i is the **error term**: everything affecting consumption besides income.

Interpretation

- If $\beta_1 > 0$, higher income is associated with higher consumption.
- The **MPC** (β_1) is: “If income increases by 1 unit, consumption increases by β_1 units (on average).”

What Does OLS Do?

OLS (Ordinary Least Squares) chooses $\hat{\beta}_0$ and $\hat{\beta}_1$ to make the fitted line as close as possible to the data points.

Fitted Values and Residuals

Given estimates $\hat{\beta}_0$ and $\hat{\beta}_1$, the predicted consumption is:

$$\hat{C}_i = \hat{\beta}_0 + \hat{\beta}_1 Y_i. \quad (2)$$

The **residual** is:

$$\hat{u}_i = C_i - \hat{C}_i. \quad (3)$$

Residuals are the vertical gaps between the points and the fitted line.

Least Squares Criterion

OLS picks $\hat{\beta}_0$ and $\hat{\beta}_1$ to minimize the **Sum of Squared Residuals (SSR)**:

$$\text{SSR} = \sum_{i=1}^n \hat{u}_i^2 = \sum_{i=1}^n (C_i - \hat{\beta}_0 - \hat{\beta}_1 Y_i)^2. \quad (4)$$

Step 1: Enter Consumption and Income Data in Excel

Open Excel and create two columns:

Income (Y)	Consumption (C)
20	18
25	21
30	24
35	27
40	30
45	33
50	36
55	38

Excel Setup

1. In cell A1 type **Income**.
2. In cell B1 type **Consumption**.
3. Enter the values in rows 2–9.

Units. You can interpret the numbers as “thousand dollars” to keep the scale simple.

Step 2: Make a Scatter Plot (Graphical Illustration)

A scatter plot shows the relationship visually.

Why we plot first

Before running regression, always plot the data to see whether a linear relationship is reasonable.

Excel Steps (Scatter Plot)

1. Highlight the range A1:B9.
2. Go to **Insert** → **Charts** → **Scatter** → **Scatter with only markers**.
3. Ensure the horizontal axis is **Income** and the vertical axis is **Consumption**.

Add Axis Titles (Recommended)

1. Click the chart.
2. Use **Chart Elements** (the “+” icon in Windows) or the chart formatting menu (Mac).
3. Add **Axis Titles**:
 - Horizontal: **Income**
 - Vertical: **Consumption**

Step 3: Add a Linear Trendline (Quick Regression Line)

Excel can draw the OLS line directly on the plot using a **trendline**.

Excel Steps (Trendline + Equation + R^2)

1. Click one of the data points in the scatter plot.
2. Right-click → **Add Trendline**.
3. Choose **Linear**.
4. Check:
 - **Display Equation on chart**
 - **Display R-squared value on chart**

What the equation means

If Excel shows something like:

$$C = 6.5 + 0.583Y,$$

then:

- Intercept 6.5 = predicted consumption when income is 0.
- Slope 0.583 = MPC: when income rises by 1 unit, predicted consumption rises by 0.583 units.

Step 4: Enable the Analysis ToolPak (Regression Output Table)

To get a full regression table (coefficients, standard errors, p-values), you need the **Analysis ToolPak**.

Windows

1. **File** → **Options** → **Add-ins**.
2. At the bottom: **Manage: Excel Add-ins** → **Go**.
3. Check **Analysis ToolPak** → **OK**.

Mac

1. **Tools** → **Excel Add-ins**.
2. Check **Analysis ToolPak** → **OK**.

Step 5: Run OLS Using Data Analysis → Regression

Excel Steps (Regression Tool)

1. Go to **Data** tab.
2. Click **Data Analysis**.
3. Select **Regression** → **OK**.

You will see a dialog with boxes. Fill them in as follows:

Regression Dialog Settings

- **Input Y Range** (Outcome): B1:B9 (Consumption)
- **Input X Range** (Predictor): A1:A9 (Income)
- Check **Labels** (because row 1 has headers).
- Choose **Output Range**: e.g., D1 (or select “New Worksheet Ply”).
- Optional (recommended): check **Residuals**.

Step 6: Understand the Regression Output

Excel produces several blocks. Focus on the following.

A) Coefficients Table (Most Important)

You will see rows like:

- **Intercept**
- **Income**

and columns like:

- **Coefficient**
- **Standard Error**
- **t Stat**
- **P-value**

Coefficient (Interpretation).

- Intercept coefficient $\hat{\beta}_0$: predicted consumption when income is 0.
- Income coefficient $\hat{\beta}_1$: predicted change in consumption for a 1-unit increase in income (the MPC).

Standard Error (Uncertainty). A standard error measures how uncertain the coefficient estimate is. Smaller standard error means more precise estimate.

t-statistic.

$$t = \frac{\text{Coefficient}}{\text{Standard Error}}.$$

Bigger $|t|$ means stronger evidence the coefficient is different from zero.

p-value. The p-value answers:

If the true effect were 0, how likely would we see a coefficient this large (in absolute value) just by chance?

Rule of thumb:

If p-value < 0.05 , we say the coefficient is statistically significant.

B) R^2 (Goodness of Fit)

R^2 is the fraction of variation in consumption explained by income (in a linear way).

$$0 \leq R^2 \leq 1.$$

Interpretation examples:

- $R^2 = 0.80$: income explains 80% of the variation in consumption.
- $R^2 = 0.10$: income explains only 10% (weak linear fit).

C) “Significance F” (Overall Test)

In a one-predictor model, this is closely related to testing whether the slope is zero. A small value (e.g., < 0.05) suggests the regression model explains consumption better than a model with no predictor.

Step 7: Compute Predicted Consumption and Residuals in Excel

Even if Excel outputs residuals, you should know how to compute them.

A) Predicted Consumption

Suppose your regression output shows:

$$\hat{\beta}_0 = (\text{Intercept}), \quad \hat{\beta}_1 = (\text{Income coefficient}).$$

Store them in two cells. For example:

- Put $\hat{\beta}_0$ in cell E2
- Put $\hat{\beta}_1$ in cell E3

(Or just reference the cells where Excel printed them.)

In column C:

- C1: Predicted_C
- C2: type the formula:

Excel Formula: Predicted Consumption

= \$E\$2 + \$E\$3 * A2

Copy the formula down to C9.

B) Residuals

In column D:

- D1: Residual
- D2: type:

Excel Formula: Residual

= B2 - C2

Copy the formula down to D9.

Step 8: Make the Residual Plot (Model Check)

A residual plot checks whether a straight line is a reasonable summary.

Excel Steps (Residual Plot)

1. Highlight A1:A9 (Income).
2. Hold Ctrl (Windows) or Cmd (Mac) and also select D1:D9 (Residuals).
3. Insert → Scatter (only markers).
4. Title the chart: **Residuals vs Income**.

How to interpret the residual plot

- **Good sign:** residuals look randomly scattered around 0.
- **Warning sign:** a curve pattern (suggests nonlinearity).
- **Warning sign:** “fan shape” (spread increasing with income) suggests unequal variance.

Alternative Method: Regression Using Excel Functions (No Tool-Pak)

Excel can compute slope, intercept, and R^2 using built-in functions.

Slope (MPC)

Excel Function: SLOPE

=SLOPE(B2:B9, A2:A9)

Intercept (Autonomous Consumption)

Excel Function: INTERCEPT

=INTERCEPT(B2:B9, A2:A9)

R^2

Excel Function: RSQ

=RSQ(B2:B9, A2:A9)

How to Write the Final Answer (Economic Interpretation)

Once you have estimates, write a sentence like:

Interpretation Template

The estimated consumption function is $\hat{C} = \hat{\beta}_0 + \hat{\beta}_1 Y$. The slope $\hat{\beta}_1$ is the MPC: a one-unit increase in income is associated with an average increase of $\hat{\beta}_1$ units in consumption. The intercept $\hat{\beta}_0$ is autonomous consumption (predicted consumption when income is zero). The R^2 indicates how much of the variation in consumption is explained by income.

Summary

- Regression fits a line: $C = \beta_0 + \beta_1 Y + u$.
- OLS chooses coefficients to minimize squared residuals.
- $\hat{\beta}_1$ (slope) is the MPC.
- Always: (1) scatter plot, (2) regression output, (3) residual plot.